



NATIONAL UNIVERSITY OF MODERN LANGUAGES

Computer Networks Assignment No - 3

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Part A: Wireless Hidden and Exposed Node Problems

1. Hidden Node Problem (with example)

The hidden node problem occurs in wireless networks when two devices (nodes) are **out of each other's transmission range** but both are within the range of a common receiver. Because they cannot detect each other's signals, they may transmit at the same time, causing a collision at the receiver.

Example Scenario:

Node A and Node C cannot hear each other, but both can hear Node B. When A and C send data to B at the same time, they collide at B.

(You can draw this simple diagram in MS Word:)

A —> B <— C

(A cannot hear C, and C cannot hear A)

2. How the Hidden Node Problem Affects Performance

- It increases collisions at the receiver.
- Causes frequent retransmissions.
- Reduces throughput and wastes bandwidth.
- Increases delay due to repeated backoff and retry attempts.

3. Causes of the Hidden Node Problem

- Physical obstacles (buildings, walls) blocking radio signals.
- Limited transmission range of wireless devices.
- Nodes located at the edges of the network.
- Interference from other nearby wireless networks.
- Non-uniform distribution of nodes in a large area.

4. Techniques to Reduce or Eliminate the Hidden Node Problem

a. RTS/CTS Mechanism:

Nodes send a Request-to-Send (RTS) before transmitting. The receiver replies with Clear-to-Send (CTS). Hidden nodes hearing CTS remain silent.

b. Increasing Transmission Power:

Expands the coverage area so nodes can sense each other.

c. Removing Physical Obstacles:

Reduces signal blockage.

d. Using Directional Antennas:

Helps focus signals and reduce interference.

e. Proper Access Point Placement:

Improves network coverage to reduce hidden nodes.

5. Exposed Node Problem (with example)

The exposed node problem occurs when a node unnecessarily stops transmitting because it hears another transmission that would not actually cause interference.

Example Scenario:

Node B is sending to A, and C wants to send to D. C hears B's transmission and assumes it must stay silent, even though sending to D would **not** cause interference.

A <— B C —> D

(C avoids transmitting even though it is safe.)

6. How the Exposed Node Problem Reduces Throughput

- Nodes stay unnecessarily silent, causing unused channel time.
- Reduces the number of simultaneous transmissions.
- Lowers overall network utilization and throughput.
- Creates inefficiency even when no real collision risk exists.

7. Comparison: Hidden Node vs Exposed Node

Aspect	Hidden Node	Exposed Node
Definition	Nodes cannot hear each other and cause collisions.	Node hears another transmission and stops unnecessarily.
Effect	Collisions increase.	Network becomes under-utilized.
Cause	Nodes out of range.	Nodes are within range but transmissions would not interfere.

Aspect	Hidden Node	Exposed Node
Result	Throughput decreases due to collisions.	Throughput decreases due to silence.

8. How RTS/CTS Helps Control Hidden and Exposed Node Problems

RTS/CTS allows nodes to check whether the channel is safe before transmission.

- For **hidden nodes**, CTS frames inform all nearby nodes to stay silent.
- For **exposed nodes**, hearing RTS/CTS allows them to understand whether they can transmit safely.

Thus RTS/CTS reduces unnecessary silence and avoids collisions.

Part B: Controlled Access Protocols

(Reservation, Polling, Token Passing)

9. What Are Controlled Access Protocols? Why Are They Needed?

Controlled access protocols allow devices to use the communication channel **one at a time**, avoiding collisions. They are needed when:

- The network has many devices.
- Collisions must be minimized.
- Fair access to the medium is required.

These protocols schedule transmissions instead of letting devices compete randomly.

10. Reservation Access Method (with explanation)

In the reservation method, time is divided into **reservation periods** and **data periods**.

- During the reservation period, each node announces (“reserves”) a time slot.
- During the data period, nodes that reserved slots transmit in their assigned order.

This ensures planned, collision-free transmission.

(You may draw a simple timeline diagram in Word showing Reservation Slot → Data Slot.)

11. Advantages and Disadvantages of Reservation Method

Advantages:

- No collisions during data transmission.
- Highly efficient under heavy traffic.
- Provides predictable access and fairness.

Disadvantages:

- Reservation overhead increases delay.
- Not efficient for light traffic.
- Requires synchronization of nodes.

12. Polling Protocol and Central Controller Role

In polling, a **central controller** (master) decides which device can transmit.

- The controller sends a “poll” signal to one device at a time.
- Only the polled device is allowed to transmit.
- After it finishes, the controller polls the next device.

This method prevents collisions and ensures orderly communication.

13. Limitations and Challenges of Polling in Large Networks

- High delay because each device must wait for its turn.
- Heavy load on the central controller (single point of failure).
- Not scalable for large networks with hundreds of nodes.
- Controller failure stops the entire network.

14. Token Passing Protocol and Token Circulation

In token passing, a special frame called a **token** circulates among nodes in a logical ring.

- A node can only transmit when it receives the token.
- After sending data, the node forwards the token to the next node.
- This ensures orderly and collision-free transmission.

15. Why Token Loss Is Critical in Token Passing Networks

The token is required for any node to start transmission. If the token is lost:

- The entire network stops transmitting.

- A new token must be regenerated, which requires time.
- If token regeneration fails, the network becomes non-operational.

16. Comparison of Reservation, Polling, and Token Passing

Feature	Reservation	Polling	Token Passing
Delay	Low after reservation	High in large networks	Moderate
Efficiency	High	Medium	High
Fairness	Good	Good	Very Good
Collision	None during data	None	None
Best for	Heavy traffic networks	Small networks	Reliable ring networks

17. Real-World Applications

Reservation:

- Satellite communication systems
- Cable TV upstream channels
- TDMA cellular networks

Polling:

- Industrial networks with a central controller
- Point-of-Sale (POS) terminals
- Old mainframe–terminal systems

Token Passing:

- Token Ring networks
- Factory automation systems
- Some real-time industrial control systems